

Module 7

Laplace Transform

Exercise Sheet

Section A – Fundamental Concepts

1. Define the Laplace Transform.

- Explain why it is used in signal and system analysis.
- How is it different from the Fourier Transform?

2. Explain the Region of Convergence (ROC).

For each of the following signals:

- $e^{-2t} u(t)$
- $e^{-3t} u(-t)$
- $e^{-2|t|}$

Find the ROC and explain its significance.

3. Determine whether the Laplace Transform exists for the following signals.

(a) $e^{-3t} u(t)$

(b) $e^{2t} u(t)$

(c) $t^2 u(t)$

(d) $\sin(5t) u(t)$

Justify your answers.

4. Find the Laplace Transform and ROC of:

(a) $e^{-2t} u(t)$

(b) $3e^{-4t} u(t)$

(c) $t u(t)$

(d) $\cos(4t) u(t)$

5. Compute the Laplace Transform of a rectangular pulse

$$x(t) = \begin{cases} 1, & 0 < t < 2 \\ 0, & \text{otherwise} \end{cases}$$

Plot the signal using Python.

Section B – Properties of Laplace Transform

6. Verify the Linearity Property.

Find the Laplace Transform of

$$3e^{-t} u(t) + 2e^{-3t} u(t)$$

using

- analytical calculation
- property of linearity

7. Verify the Time-Shifting Property.

Given

$$x(t) = e^{-t} u(t)$$

find the Laplace Transform of

$$x(t - 2)$$

Explain how the exponential factor changes.

8. Verify the Frequency-Shifting Property.

Given $x(t) = \cos(3t) u(t)$

find the Laplace Transform of

$$e^{-2t} x(t)$$

Compare the poles before and after shifting.

9. Verify the Differentiation Property.

For $x(t) = te^{-2t} u(t)$

Find the Laplace Transform directly and using the differentiation property.

10. Initial and Final Value Theorem

For each system below,

(a) $X(s) = \frac{5}{s+2}$

(b) $X(s) = \frac{4}{s(s+1)}$

Determine

- Initial value
- Final value

Verify using inverse Laplace Transform.

Section C – Programming Problems

11. Compute Laplace Transform using Python

Write a Python program using **SymPy** to compute the Laplace Transform of

- e^{-2t}
- $\sin(4t)$

- t^2

Compare the symbolic results with analytical calculations.

12. Pole-Zero Analysis

For the transfer function

$$H(s) = \frac{s+2}{(s+1)(s+4)}$$

- Find poles.
- Find zeros.
- Plot the pole-zero diagram.
- Comment on stability.

13. Solve a Differential Equation

Solve $y'(t) + 3y(t) = 5$, $y(0) = 2$

using

- Laplace Transform
- Plot the solution.

14. Verify the Inverse Laplace Transform

Find the inverse Laplace Transform of

$$\frac{s+4}{(s+2)(s+3)}$$

Verify the result using Python.

15. Compare Two Systems

Consider $H_1(s) = \frac{1}{s+1}$

and $H_2(s) = \frac{1}{s+5}$

Plot their impulse responses and explain

- Which system is faster?

- Which decays more rapidly?
 - How do the pole locations affect the response?
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Challenge Problems

Challenge 1

The Laplace Transform $\frac{1}{s+2}$

can represent different time-domain signals depending on the ROC.

Find all possible signals and explain why the ROC changes the interpretation.

Challenge 2

Build a **Laplace Transform Calculator** using SymPy that accepts a user-defined function and computes its Laplace Transform automatically.

Challenge 3

Generate pole-zero plots for at least five transfer functions and classify each system as:

- Stable
- Unstable
- Marginally Stable

Provide justification based on pole locations.

Challenge 4

Analyze an RC Circuit using the Laplace Transform.

- Derive the transfer function.
- Find poles.
- Plot the step response.
- Explain the circuit behavior.

Challenge 5

Solve the same differential equation using:

- Classical calculus
- Laplace Transform
- Python (SymPy)

Compare all three methods.